

ELECTIVE COURSE-8.1

UEC410: REAL-TIME OPERATING SYSTEM				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: The purpose of this course is to provide students an in depth understanding of real-time operating systems, real-time debugging, and the use of RTOS in embedded systems.				
Introduction to Real Time Operating System: RTOS vs GPOS, Task Scheduling, Latency, Priority Inversion, Types of RTOS kernels.				
Multitasking: Task Model, Task templates, States of Task, Application partitioning, Assigning Priorities, Task Scheduling, Task Services, Shared Resources, Inter-task Access synchronization.				
Thread Management: Types of threads, Thread Control Block, Creation of Threads, Switching Threads, Profiling, Thread synchronization, Process Management, Concept of FIFO.				
Real Time Systems: Embedded System with ARM processor, ARM Bus Technology and AMBA Bus Protocol, Memory, Peripherals, Cache and Tight Couple memory, Coprocessor Extensions, Porting RTOS on ARM device, Basic RTOS tasks using ARM processor				
Course Learning Outcomes (CLOs) The students will be able to: 1. understand real-time operating systems, real-time debugging, and embedded systems 2. design real-time embedded systems, such as motor controllers and robotic systems. 3. build task schedulers and setting up priorities. 4. design ARM based real times systems by performing tasks using RTOS.				
Text Books 1. Embedded Systems: Real-Time Operating Systems for ARM® Cortex™-M Microcontrollers, Jonathan W. Valvano, Create Space Independent Publishing Platform, 5th ed., 2019. 2. Operating Systems: Principles and Practice, Thomas Anderson and Michael Dahlin, Lightning Source Inc, 2014.				
Reference Books 1. The Definitive Guide to the ARM Cortex-M3 and Cortex-M4 Processors, Joseph Yiu, Newnes, 3rd ed., 2013.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST	35
2.	EST	35
3.	Sessional (Quizzes/Assignments/Lab Evaluations)	30

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